

BIOGRAPHICAL SKETCH

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NAME: Kissler, Stephen

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POSITION TITLE: Assistant Professor of Computer Science

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
University of Colorado Boulder, Boulder, CO	BS	05/2013	Applied Mathematics
University of Colorado Boulder, Boulder, CO	MS	05/2014	Applied Mathematics
University of Cambridge, Cambridge	PHD	05/2018	Applied Mathematics
University of Cambridge, Cambridge	Fellow	01/2019	
Harvard T.H. Chan School of Public Health, Boston, MA	Postdoctoral Fellow	08/2023	

A. Personal Statement

My research is dedicated to improving preparedness for infectious disease outbreaks. I develop mathematical models and digital tools to guide public health policy. I have a broad background in mathematics and infectious disease epidemiology and have used this expertise to characterize the dynamics of respiratory viral infection within the body, to explain drivers of disease transmission within and between communities, and to participate in international pandemic response (1-4). I built my foundation in mathematical epidemiology as a Gates Scholar at the University of Cambridge, where I identified social and demographic determinants of 2009 A/H1N1 pandemic influenza transmission. As a postdoctoral fellow at the Harvard T.H. Chan School of Public Health, I used a mix of data science, statistical inference, and mechanistic modeling to identify drivers of antibiotic prescribing, to infer optimal vaccination strategies, and to build situational awareness during the COVID-19 pandemic.

Since beginning my faculty position in 2023, I have completed various research projects, including an assessment of SARS-CoV-2 viral kinetics in individuals with repeat infections (3) and an estimate of the combined impact of vaccination and behavior on reducing COVID-19 mortality in the United States (4), supported by startup funds from the University of Colorado Boulder. Ongoing projects include an assessment of the impact of emerging respiratory viruses on food supply chains (funded by university startup, with NIH DP2 grant pending); the design of contact tracing studies to infer the timing and intensity of infectiousness (funded by university startup); the design of optimal scoring rules for epidemiological forecasting competitions (funded by university startup, with University of Colorado seed grant pending); and the design of diagnostic test-based policies for controlling respiratory viral disease outbreaks, including the creation of a public viral kinetic database (funded by the National Science Foundation and the Centers for Disease Control and Prevention). On completed projects (3) and (4), I was a co-investigator, leading the mathematical analysis and assisting with project coordination. On the remaining ongoing projects, I am the lead investigator, overseeing all aspects of the project's completion.

I am passionate about research mentorship and currently oversee two doctoral students, three masters students, and a postdoctoral candidate. I work collaboratively with scientists whose expertise ranges from clinical medicine to economics to behavioral science. Additional service activities over the past five years

include serving as a reviewer for various grants (NSF "Integrating human behavior in epidemiological models", 2023; Marsden Fund, 2024); hosting a podcast on COVID-19 epidemiology (101 episodes from 2020-2022); serving as a member of the World Health Organization Access to COVID-19 Tools Accelerator (ACT-A) modeling consortium (2020-21) and the CDC COVID-19 modeling working group (2021-23); and coordinating a peer observation program for pedagogical improvement (2024-present); along with ongoing committee service (Graduate student mentorship committee; BioFrontiers Institute council) and peer reviews of scientific manuscripts.

Building upon my experience and professional ties, I aim to support public health surveillance and response by building digital infrastructure to synthesize complex data sources and provide equitable, community-tailored guidance for infectious disease control.

1. Atkeson A, Kissler S. The Impact of Vaccines and Behavior on US Cumulative Deaths from COVID-19. Brookings Papers on Economic Activity. 2024 March 14. Available from: https://www.brookings.edu/wp-content/uploads/2024/03/2_Atkeson-Kissler_unembargoed.pdf
2. Kissler SM, Hay JA, Fauver JR, Mack C, Tai CG, Anderson DJ, Ho DD, Grubaugh ND, Grad YH. Viral kinetics of sequential SARS-CoV-2 infections. Nat Commun. 2023 Oct 5;14(1):6206. PubMed Central PMCID: PMC10556125.
3. Kissler SM, Grad YH. ANTICIPATING RACIAL/ETHNIC MORTALITY DISPLACEMENT FROM COVID-19. Am J Epidemiol. 2022 Jul 23;191(8):1519-1520. PubMed Central PMCID: PMC9383750.
4. Kissler SM, Mitchell M, Grad YH. Reduction in Antibiotic Prescribing Attainable With a Gonococcal Vaccine. Clin Infect Dis. 2021 Sep 15;73(6):e1368-e1371. PubMed Central PMCID: PMC8522794.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2023 -	Assistant Professor of Computer Science, University of Colorado Boulder, Boulder, CO
2020 - 2020	Teaching Fellow, Harvard T.H. Chan School of Public Health, Boston, MA
2019 - 2023	Postdoctoral Research Fellow, Harvard T.H. Chan School of Public Health, Boston, MA
2018 - 2019	Junior Research Fellow, University of Cambridge, Cambridge
2015 - 2018	Supervisor, Mathematical Biology, University of Cambridge, Cambridge

Honors

2018 - 2019	Junior Research Fellowship, Queens' College Cambridge
2014 - 2018	Gates Cambridge Scholarship, Gates Cambridge Trust
2009 - 2014	Boettcher Scholarship, Boettcher Foundation
2022	NOMIS & Science Young Explorer Award Finalist, NOMIS Foundation and AAAS
2022	Barry R. and Irene Tilenius Bloom Fellowship, Harvard T.H. Chan School of Public Health
2018	Institut Mittag-Leffler Workshop Fellowship, Institut Mittag-Leffler
2015	65th Lindau Nobel Laureate Meeting and Fellowship, Bayer Science and Education Foundation
2014	M.S./B.S., Summa Cum Laude with honors, University of Colorado Boulder
2014	Outstanding Graduate, College of Engineering, University of Colorado Boulder

C. Contribution to Science

1. Identification of geographic and demographic drivers of pandemic influenza transmission

In an emerging respiratory viral pandemic, it is critical to anticipate how infection will spread, both geographically and across demographic groups. Experience from various endemic respiratory

pathogens may lead one to expect that early pandemic transmission may center on highly connected cities, and that children may contribute most to transmission due to transmission in schools. However, a lack of finely resolved case data had prevented us from assessing these hypotheses rigorously. Using medical claims data from 2009-10, we extracted outpatient visits for influenza-like illness at the ZIP code level in the United States and used them to map the spread of the 2009 A/H1N1pdm influenza pandemic. We built a mechanistic model to explain how the epidemic jumped between cities, incorporating various predictors including school start dates, geographic proximity, and mixing. We then developed a novel technique to trace chains of transmission back to their starting points. We found that these early focal points of transmission were generally small, outlying cities, not major travel centers. Furthermore, the epidemic radiated from these foci in a coherent wave, rather than jumping from city to city, suggesting that air travel was less important for the geographic spread of influenza within the United States than had been previously thought. This led us to hypothesize that children might be most responsible for spread, since children do not tend to travel as far or as frequently as adults. We developed a novel signal processing approach to infer which age groups drive transmission in which other age groups, based on time series data. We found that school-aged children were indeed responsible for the bulk of transmission. Taken together, these findings suggest that school-aged children are an important population for transmission-reducing interventions during an influenza pandemic, and that concentrating early surveillance and response efforts in major cities may not be the most efficient way to quell an emerging influenza pandemic in the United States. The methods developed during this work can be easily translated to other pathogens as well.

- a. Kissler SM, Viboud C, Grenfell BT, Gog JR. Symbolic transfer entropy reveals the age structure of pandemic influenza transmission from high-volume influenza-like illness data. *J R Soc Interface*. 2020 Mar;17(164):20190628. PubMed Central PMCID: PMC7115222.
- b. Kissler SM, Gog JR, Viboud C, Charu V, Bjørnstad ON, Simonsen L, Grenfell BT. Geographic transmission hubs of the 2009 influenza pandemic in the United States. *Epidemics*. 2019 Mar;26:86-94. PubMed PMID: 30327253.

2. Collection of a large human mobility and social contact dataset

A key barrier to anticipating the geographic spread of an epidemic is the lack of data on how individuals move and interact with one another. To date, most geographic disease transmission studies have relied on travel proxies, like commuting data or air traffic flows, or theoretical mobility models. Recent studies have turned to mobile phone geolocation data, but due to privacy concerns these data are not available at the individual level, which is the scale at which disease transmission occurs. To address this gap, we collaborated with the British Broadcasting Company (BBC) to recruit 48,000 volunteers from the UK to participate in the largest joint human mobility and social contact study to date. Volunteers downloaded a mobile phone app that tracked their movements hourly for 24 hours and then prompted them to report their interpersonal interactions from that day. We also conducted a more targeted study in the town of Haslemere that tracked volunteers' movements continuously for three days. Using the data, we simulated how an outbreak of an influenza-like respiratory pathogen might spread in the UK and ranked a variety of targeted vaccination strategies in terms of their effectiveness in reducing the intensity and final size of the simulated outbreak. These data were used in the UK to guide policy during the COVID-19 pandemic.

- a. Kissler S, Klepac P, Tang M, Conlan A, Gog J. Sparking "The BBC Four Pandemic": Leveraging citizen science and mobile phones to model the spread of disease. *BioRxiv*. 2018; :479154.

3. Antibiotic prescribing trends

Antibiotic resistance is a major threat to global health. Antibiotic resistance is driven principally by antibiotic consumption, motivating the need to understand the drivers of antibiotic consumption. Previous efforts in this area have focused on the point of care, assessing the patient- and provider-associated reasons for inappropriate antibiotic prescribing (e.g., antibiotic prescriptions for non-

bacterial infections). Still, even appropriately prescribed antibiotics may contribute to resistance, suggesting that upstream interventions, like infection prevention through vaccination and non-pharmaceutical interventions, may also be an effective way to reduce antibiotic prescribing and associated resistance. To guide such interventions, it is critical to know which medical conditions drive antibiotic prescribing, in which populations, and for what reasons. However, a lack of data linking pharmacy claims, patient-provider encounters, and patient demographic information were a major barrier to gaining these insights. We addressed this challenge by analyzing various high-volume medical insurance claims databases and linking them with epidemiological survey data and demographic information from the Centers for Disease Control and Prevention and the US Census Bureau. We found that antibiotic prescribing in the United States is driven principally by respiratory infections in children, and that the volume of antibiotic prescribing closely follows socioeconomic status, such that individuals with barriers to care are less likely to receive antibiotics. This suggests that, alongside over-prescribing in some groups, there may also exist an under-prescribing problem that must be addressed to ensure that interventions aimed at reducing antibiotic prescribing do not divert critical treatments away from populations that already face high barriers to care. Together, these findings indicate that antibiotic stewardship and health equity must be addressed in tandem, and that reducing respiratory infection in children may be an effective, equitable way of reducing antibiotic prescribing and associated resistance.

- a. Kissler SM, Wang B, Mehrotra A, Barnett M, Grad YH. Impact of Respiratory Infection and Chronic Comorbidities on Early Pediatric Antibiotic Dispensing in the United States. *Clin Infect Dis.* 2023 Feb 8;76(3):382-388. PubMed Central PMCID: PMC9907510.
- b. Kissler SM, Klevens RM, Barnett ML, Grad YH. Childhood Respiratory Outpatient Visits Correlate With Socioeconomic Status and Drive Geographic Patterns in Antibiotic Prescribing. *J Infect Dis.* 2021 Jun 15;223(12):2029-2037. PubMed PMID: 33903899.
- c. Kissler SM, Klevens RM, Barnett ML, Grad YH. Distinguishing the Roles of Antibiotic Stewardship and Reductions in Outpatient Visits in Generating a 5-Year Decline in Antibiotic Prescribing. *Clin Infect Dis.* 2021 May 4;72(9):1568-1576. PubMed PMID: 32173727.

4. SARS-CoV-2 situational awareness

As SARS-CoV-2 began spreading around the world, there was an urgent need to assess how the pandemic might unfold in the coming months and years and the degree of physical distancing that could be required to mitigate its worst effects. We used epidemiological data on existing human coronaviruses and the most recent data on SARS-CoV-2 to address pressing questions about its spread, including whether transmission would be possible during the summer months, under which circumstances the virus could enter into seasonal circulation, and for how long intermittent physical distancing might be necessary to keep surges from overwhelming hospital capacities in the absence of other interventions. As the pandemic continued, much of the burden of disease and mortality was concentrated in socioeconomically disadvantaged individuals. To effectively intervene, it was necessary to understand to what extent the difference in disease burden was due to differences in underlying health and quality of care after infection, and to what extent it was due to different rates of infection across these groups, since each problem would imply different ways of intervening. Using viral testing data from New York City, we assessed differences in SARS-CoV-2 prevalence across boroughs and linked these differences with mobility patterns. We found that prevalence was reliably highest in areas where individuals commuted out of the borough in the morning and back in the evening, consistent with work patterns, suggesting that high infection rates correlated with workplace exposure. This work helped to advocate for greater infection prevention measures for essential workers.

- a. Kissler SM, Kishore N, Prabhu M, Goffman D, Beilin Y, Landau R, Gyamfi-Bannerman C, Bateman BT, Snyder J, Razavi AS, Katz D, Gal J, Bianco A, Stone J, Larremore D, Buckee CO, Grad YH. Reductions in commuting mobility correlate with geographic differences in SARS-CoV-2

prevalence in New York City. *Nat Commun.* 2020 Sep 16;11(1):4674. PubMed Central PMCID: PMC7494926.

- b. Kissler SM, Tedijanto C, Goldstein E, Grad YH, Lipsitch M. Projecting the transmission dynamics of SARS-CoV-2 through the postpandemic period. *Science.* 2020 May 22;368(6493):860-868. PubMed Central PMCID: PMC7164482.

5. SARS-CoV-2 viral kinetics

The trajectory of viral abundance in the body during an acute infection (i.e., viral kinetics) underlies many important facets of infection transmission and control, including infectiousness over time and test sensitivity. Furthermore, variation in viral kinetics across population groups can help to reveal important correlates of immune response. However, measuring viral kinetics has historically been challenging, requiring frequent, longitudinal quantitative PCR testing, which is costly. To address this challenge, we collaborated with the National Basketball Association's (NBA's) occupational health program, where routine quantitative PCR testing was administered for infection control purposes. We developed a mathematical model to extract differences in mean viral kinetics across population groups. We found that immunity, either through previous infection or vaccination, tends to reliably reduce the duration of viral shedding. This shortening of acute infection may partially underlie the reduced infectiousness of vaccinated and previously infected individuals. Furthermore, in individuals who were infected twice, we found that those who cleared their first infections quickly also cleared their second infections quickly, and vice versa. This points towards some element of the immune response that persists across infections, such as an inherent, genetically mediated goodness of response, or possibly a lasting immune memory of prior exposure to related, non-SARS coronaviruses. Together, this work helps to clarify the immune mechanisms involved in SARS-CoV-2 infection and provides a framework for analyzing viral kinetics in acute infections applicable to a wide range of respiratory pathogens.

- a. Hay JA, Kissler SM, Fauver JR, Mack C, Tai CG, Samant RM, Connolly S, Anderson DJ, Khullar G, MacKay M, Patel M, Kelly S, Manhertz A, Eiter I, Salgado D, Baker T, Howard B, Dudley JT, Mason CE, Nair M, Huang Y, DiFiori J, Ho DD, Grubaugh ND, Grad YH. Quantifying the impact of immune history and variant on SARS-CoV-2 viral kinetics and infection rebound: A retrospective cohort study. *Elife.* 2022 Nov 16;11 PubMed Central PMCID: PMC9711520.
- b. Kissler SM, Fauver JR, Mack C, Tai CG, Breban MI, Watkins AE, Samant RM, Anderson DJ, Metti J, Khullar G, Baits R, MacKay M, Salgado D, Baker T, Dudley JT, Mason CE, Ho DD, Grubaugh ND, Grad YH. Viral Dynamics of SARS-CoV-2 Variants in Vaccinated and Unvaccinated Persons. *N Engl J Med.* 2021 Dec 23;385(26):2489-2491. PubMed Central PMCID: PMC8693673.
- c. Kissler SM, Fauver JR, Mack C, Olesen SW, Tai C, Shiue KY, Kalinich CC, Jednak S, Ott IM, Vogels CBF, Wohlgemuth J, Weisberger J, DiFiori J, Anderson DJ, Mancell J, Ho DD, Grubaugh ND, Grad YH. Viral dynamics of acute SARS-CoV-2 infection and applications to diagnostic and public health strategies. *PLoS Biol.* 2021 Jul;19(7):e3001333. PubMed Central PMCID: PMC8297933.